

Social Media Sentiment Network Analysis

SkilledScore — Data Visualization Internship

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Dataset & Methodology

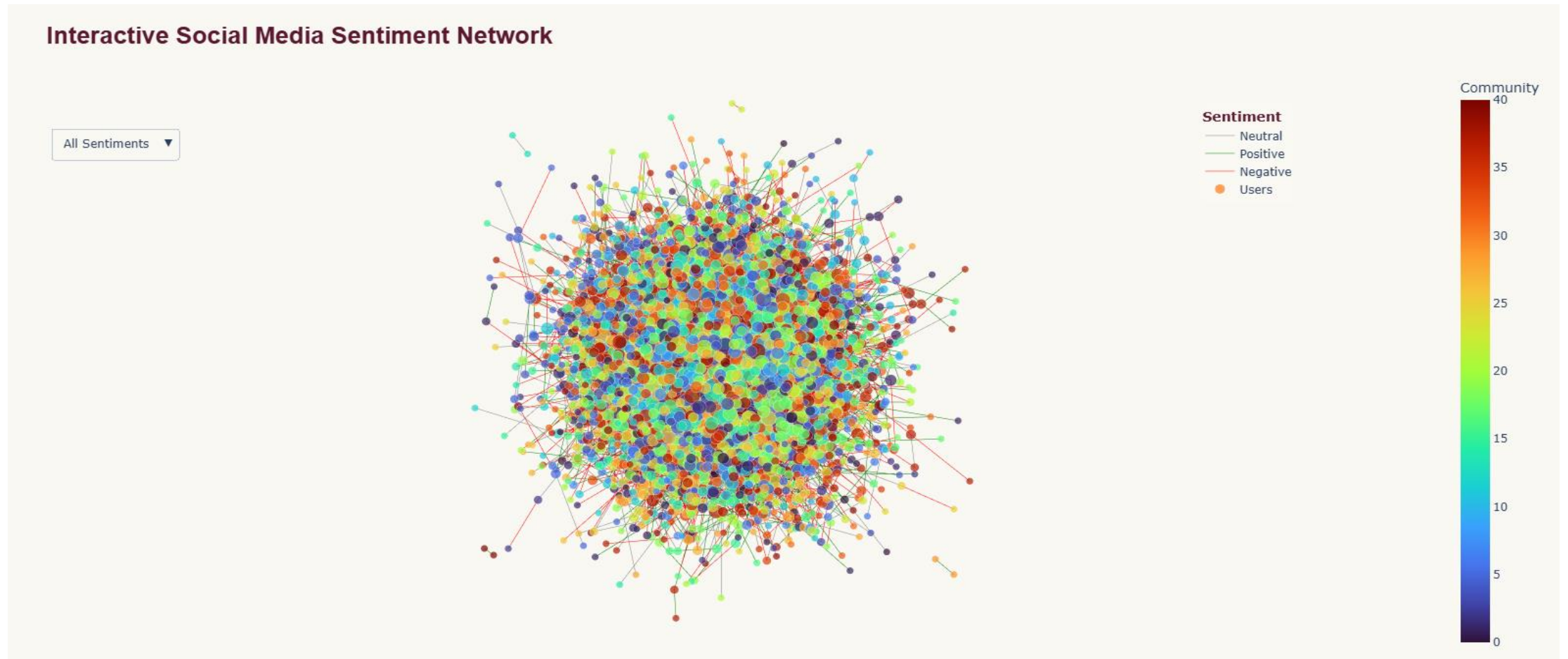
Synthetic Dataset

- 5,000 users
- 20,000 interactions: posts, retweets, and replies
- Features: User ID, Post ID, Timestamp, Sentiment Score, Interaction Type, and Hashtags

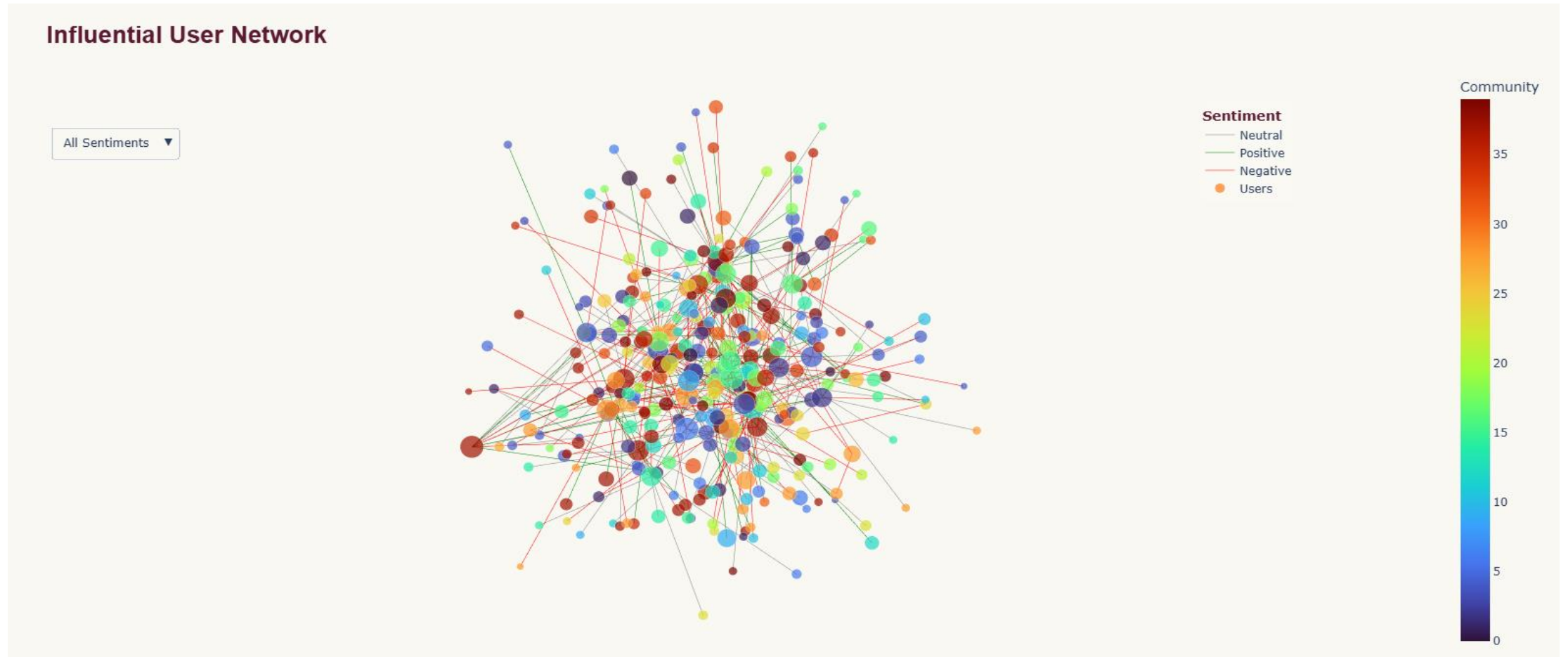
Methodology

- VADER sentiment analysis with scores ranging from -1 to +1
- Duplicate removal and low-activity user filtering
- Directed interaction network constructed using NetworkX
- Degree centrality used to measure user influence
- Louvain algorithm used for community detection
- Force-directed layout used for network visualization

Interactive Social Media Sentiment Network



Influential Users & Sentiment Insights



Key Findings & Moderation Implications

Key Findings

- Sentiment distribution: 37.12% neutral, 33.48% negative, and 29.40% positive.
- Degree centrality identified influential users with differing sentiment patterns.
- Louvain analysis identified 41 communities in the final reproducible run.
- Dominant hashtag concentration reached only 13.86% among meaningful communities.
- Larger communities showed mixed sentiment and limited evidence of strong hashtag-driven echo chambers.

Moderation Implications

- Prioritize high-influence users and interaction-level patterns for monitoring.
- Use sentiment as a screening signal rather than treating negative sentiment as a policy violation.
- Community and hashtag membership should provide context rather than determine moderation decisions.

Note: All results are based on synthetic data.